

Multilevel fusing paired visible light and near-infrared spectral images for face anti-spoofing

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ABSTRACT

Face anti-spoofing has become a vital element for guaranteeing the security of face recognition systems. Previous face anti-spoofing approaches generally exploit cues in visible light images or near-infrared images individually. Few studies pay attention to fusing visible light and near-infrared images for face anti-spoofing. However, the strengths and weaknesses of visible light images and near-infrared images for face anti-spoofing can be complementary. In this study, we introduce a new face anti-spoofing dataset named as CIGIT-PPM, which includes paired visible light and near-infrared images with spoofing medium, distance, pose, expression and session variations for both print and 3D mask attacks. Further, we propose a novel fusing paired visible light and near-infrared spectral images CNN based approach for face anti-spoofing, combining the discriminating ability of both visible light and near-infrared spectral images. Specifically, an end-to-end CNN is employed to learn fusing representation from paired images and do classification at multilevel, and then weight averaged strategy is utilized to integrate the classification probability of different levels, which gives the final result that the input paired images are captured from a live face or a spoof face. Extensive experiments show that our method achieves state-of-the-art results on self-collected dataset CIGIT-PPM and public dataset msspoof.

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1. Introduction

Face recognition is wildly used in various business scenarios such as identity check in public transport, electronic payment and entrance guard system. To some extent, face images gradually become a representation of identity. However, conventional face recognition systems have a vulnerability in security because they do not distinguish the input face images are captured from a live face or a spoof face. Aiming at this vulnerability, three common spoof faces including faces printed on paper, face videos replayed on digital devices, and 3D face masks [17] have gradually emerged to fool face recognition systems.

To improve the security of face recognition system, a lot of face anti-spoofing methods have been proposed in recent years. Most of methods regard face anti-spoofing as a classification task based on the images captured under visible spectrum (VIS images). The differences in texture [4,5,11], image quality [8,23], motion [3,10,13,19], vital signs [12,21] are utilized as cues to distinguish

live faces from spoof faces. However, these methods are usually vulnerable to illumination, image quality, image acquisition equipment, and materials of spoof faces. The VIS images captured from high quality spoof faces look similar to the VIS images captured from live faces, as shown in Fig. 1, so it is difficult to find effective and generalized cues to classify the live and spoof faces only using VIS images. Fortunately, the materials used to make spoof faces like paper, screen of digital devices, latex, silicone usually have different reflection properties with live faces under near-infrared spectrum. As shown in Fig. 1, there are obvious differences between the images of live and spoof faces captured under near-infrared spectrum (NIR images). For instance, the skin of live faces looks more smooth and delicate compared to spoof faces. Fake eyes look clearer than real eyes. Sun et al. [20,26] have taken these reflection differences as classification cues and proved that these differences have an advantage in improving performance of face anti-spoofing. However, NIR images are not as rich as VIS images in color and texture information, so NIR images based methods also lose many effective cues which exist in VIS images.

Following this perspective, the strengths of VIS and NIR images are different but complementary. Therefore, in this paper we propose to fuse the cues preserved in VIS and NIR images for face

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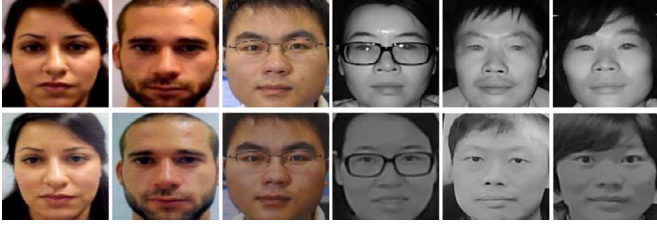


Fig. 1. Comparisons of live and spoof face images captured under visible light and near-infrared spectrum. The first three columns are VIS images and the other three are NIR images. The first row are live face images and the second row are spoof face images.

anti-spoofing. Considering the effectiveness of convolution neural network in representations learning and classification, we utilize CNN to learn fusing features and classify live and spoof faces.

As we know, face anti-spoofing research based on visible light and near-infrared spectrum requires VIS and NIR images. To the best of our knowledge, existing public face anti-spoofing dataset msspoof [6] contains paired VIS and NIR images. However, the photos used to collected data in msspoof are printed to black-and-white photos. Black-and-white photos loss many color and texture cues for face anti-spoofing. Furthermore, the msspoof dataset does not cover 3D mask attack. MLFP [1] contains real face and 3D mask attack images recorded under visible light, near-infrared and Thermal spectrum. Unfortunately, MLFP dataset only has mask attack, no photo attack. It is necessary to build a more comprehensive dataset for face anti-spoofing research based on visible light and near-infrared spectrum. Therefore, we collect a new face anti-spoofing dataset, named as CIGIT paired VIS and NIR images for photo and mask attacks (CIGIT-PPM). CIGIT-PPM dataset consists of 93,358 paired VIS and NIR face images for 72 subjects and 2 attack types, covering variations in terms of presentation attack, spoofing medium, recording environment, pose, expression, glasses/no glasses, resolution, distance. The main contributions of this work are listed as follows:

- (1) We release a new dataset which contains paired visible light and near-infrared face images with more data volume and diversity for face anti-spoofing.
- (2) We utilize convolution neural network to learn multi-level fusing features from paired VIS and NIR images for face anti-spoofing, combining the advantages of visible light and near-infrared spectrum. It is effective not only for print attack, but also for 3D mask attack.
- (3) Extensive experiments are carried out on CIGIT-PPM and msspoof dataset. We achieve the state-of-the-art results on both intra-dataset and cross-dataset testing for face anti-spoofing.

2. Related work

We review related work in an input data diversity perspective. The approaches proposed for face anti-spoofing can be divided into two categories: visual light based methods, multispectrum based methods.

Visual light based methods Most of prior studies utilize VIS images to do face anti-spoofing because they can use existing cameras of face recognition system. Some methods seek to find vital signs from VIS images such as eye blinking [13,19], lip motion [10], head rotation [3], pulse [12], and skin blood flow [21]. However, vital signs analysis need to obtain sequence information from face video. These methods cannot get result from single frame. Furthermore, most of them are sensitive to motion and illumination. Considering the mediums and color distribution of spoof faces are different from live faces, many researches [4,5,11,25] focus on color

texture analysis via LBP, HOG, DoG. They transfer VIS images from RGB color space into grey, HSV and YCbCr color spaces to get more discriminative feature. These methods get significant improvement for face anti-spoofing, but T. de Freitas Pereira et al. [7] point that they are lack of generalization ability. Their classification accuracy depends on whether the testing data is collected similarly as training data. Image quality based methods [8,23] extract image distortion information from VIS images including specular reflection, blurriness, chromatic moment and color diversity. The generalization of this method is well but it requires high quality VIS images to extract these distortion cues. Besides hand-crafted features, deep convolution neural network is also utilized to learn discriminative features from VIS images in RGB, HSV and YCbCr color spaces [2,22,24].

Multispectrum based methods Images captured under multiple spectrums cross 400–1200 nm are also utilized to do face anti-spoofing. In [20], context and lighting features are extracted from near-infrared differential images to do face anti-spoofing. Yi et al. [26] present a multispectral face recognition system and analyse the color and texture information from near-infrared and visible spectral images via HoG and Gabor features. These methods focus on using near-infrared spectral images to process print attack. The printed photos have significant reflection difference with live faces under near-infrared spectrum, but masks have less difference with live faces than printed photos. As a consequence, the performance of those methods degrades significantly while meeting 3D mask attack. Short wave infrared spectral images are used to distinguish human skin from other materials for 3D mask attack [18]. This method depends on the proportion of uncovered skin region to do face anti-spoofing. Raghavendra et al. [14] extract time-frequency features from the input multispectral face images independently via L_aMT_iF descriptors and utilize SVM to obtain the binary decision. A multispectral face anti-spoofing database including images captured under seven individual spectral bands is introduced in [16]. Spectral signatures in these multispectral images are explored. Reflectance curves formed by spectral signatures are used to distinguish live faces from spoof faces [15].

Compared to related methods, we learn multilevel fusing features from VIS and NIR images instead of VIS images or NIR images individually. We utilize CNN to learn fusing features instead of hand-crafting features. Most of the related methods concentrate on one attack type, for example print or 3D mask attack, but we deal with presentation attacks mixed with print and 3D mask attacks in a single model for face anti-spoofing.

3. Collection of the CIGIT-PPM dataset

Dataset is a critical part of face anti-spoofing research. Face anti-spoofing research based on visible light and near-infrared spectrum needs paired VIS and NIR images. To facilitate the study, we collect a new dataset named CIGIT paired VIS and NIR images for print and mask attacks (CIGIT-PPM). Table 1 shows the comparison of our collected dataset and similar existing datasets for face anti-spoofing. Our collected dataset has multiple advantages over previous datasets. First, it contains more subjects and samples, e.g., 3 times of the subjects of msspoof. Second, it contains more presentation attack types. The recorded presentation attacks of our dataset include print attack and 3D mask attack. Third, it contains more data diversity in terms of spoofing medium, recording environment, pose, expression, glasses/no glasses, resolution and distance.

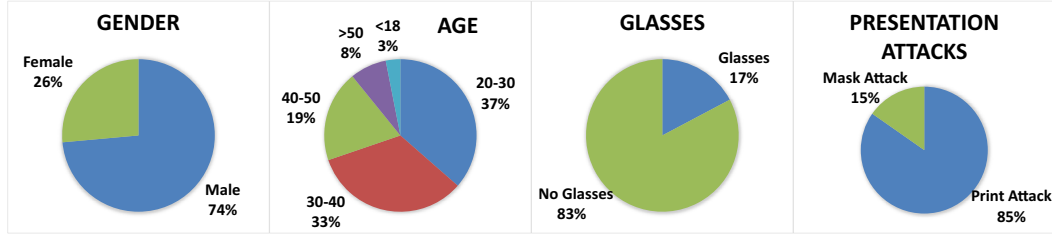
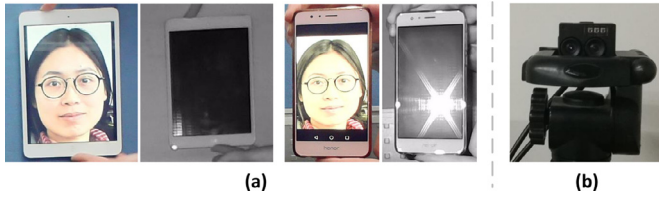
3.1. Preparation of live and spoof faces

CIGIT-PPM includes VIS and NIR face image pairs of real access and attack attempts from 72 subjects. The 72 subjects consist of

Table 1

The comparison of our collected dataset and similar existing datasets for face anti-spoofing.

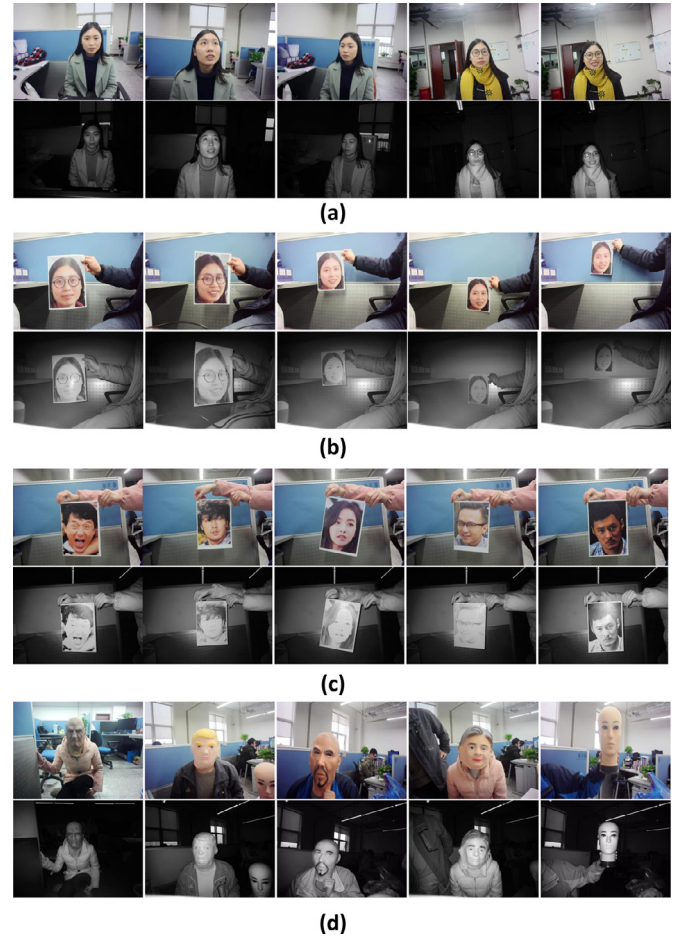
Dataset	Year	# of Subj.	# of Sample	Spoof attacks	Data variant
msspoof [6]	2016	22	2247 pair	print	session
MLFP [1]	2017	10	1350 video	3D mask	session, spoofing medium
CIGIT-PPM	2018	72	93,358 pair	print, 3D mask	session, spoofing medium, pose, expression, distance, glasses/no glasses, resolution

**Fig. 2.** The statistics of the subjects in the CIGIT-PPM dataset.**Fig. 3.** (a) Two example paired VIS and NIR images of display attacks using Apple iPad mini2 and Huawei Honor8, respectively. (b) The visible light and near-infrared binocular camera used to record image pairs.

61 live persons and 11 masks. 31 of these 61 live persons are staff of our institute, and the remaining 30 are celebrities. Some statistics of the subjects are shown in Fig. 2. The subjects of CIGIT-PPM are diverse in gender, age, glasses/no glasses. The presentation attacks of CIGIT-PPM include print attack of 85% of subjects and mask attack of 15% of subjects.

It can be observed that we cannot see face region from the NIR frame of display attack, as shown in Fig. 3(a). It is easy to detect display attack under near-infrared spectrum, so we do not consider display attack when we collect data. We only record data for real access, print attack and 3D mask attack attempts. We record live face image pairs from the 31 persons who work for our institute. We record spoof face image pairs from the printed photos of the 61 live subjects and 11 masks. We take three pictures for every subject of the 31 colleagues by Nikon D90 Single-lens reflex camera. The three pictures differ in pose and expression of subject from one another. We collect face pictures of the 30 celebrities from internet and pick one picture for each celebrity. Most of pictures from internet are taken by different cameramen and cameras in the wild rather than constrained experimental conditions. Utilizing the face pictures from internet is beneficial to improve the diversity of spoof faces in terms of photographic equipment and environment. Besides, it is also useful to enrich the diversity of spoof faces in terms of pose, expression and occlusion. All Pictures are printed into chromatic photos on A4 paper using a Cannon image-CLASS MF249dw printer.

11 masks are used to collect paired image data for mask attacks. Ten masks can be worn on head. One mask is a head mould. The materials of the masks are 2 leather, 8 rubber, 1 plastic. These 11 masks are made by different factories, so the manufacturing process and materials are different from each other. This character is benefit for the image data diversity of the whole dataset. The sample live and spoof face image pairs are shown in Fig. 4.

**Fig. 4.** From (a) to (d) are samples of real accesses, print attacks of our colleagues, print attacks of celebrities, 3D mask attacks, respectively. The collected VIS and NIR image pairs have session, pose, expression, spoofing medium, distance, position, and glasses/no glasses variation.

3.2. Acquisition details of CIGIT-PPM

As shown in Fig. 3(b), a visible light and near-infrared binocular camera is used to record VIS and NIR frames for real access and attack attempts. The VIS and NIR frames are captured by the binocular camera simultaneously to form the frame pair. The wavelength



Fig. 5. Samples of aligned VIS and NIR face image pairs in CIGIT-PPM dataset. The first row are VIS images. The second row are corresponding NIR images.

of the near-infrared camera is 850 nm. The resolution of the visible light camera and near-infrared camera are all two megapixels.

To enrich the data diversity, we pay attention to some details when we record image pairs. Acquisition environment is one of influencing factor of face anti-spoofing performance. Therefore, we record data in three different sessions separated by a time interval of about one month. Every session has different illumination condition and indoor scene. In session 1, the recordings are taken in the central office. The lights are turn on. In session 2, the recordings are taken near the door and windows of the office. The lights are turn on. The door and windows are behind of the recorded objects. In session 3, the recordings are taken at a corner of the office where has less lights and is darker than session 1 and session 2.

Distance between camera and recorded object influences the intensity of near-infrared light, and influences the NIR images imaging. In order to get image pairs of different distance, the recorded object (e.g. live person, photo, 3D mask) is required to move back and forth within a distance of 30–100 cm from the binocular camera when recording data. When we collect image pairs for real access attempts, the subjects are asked to change expression and the yaw and pitch angle of the head within $[-45^\circ, +45^\circ]$. As a consequence, the dataset contains image pairs having pose and expression variants. Subjects who wear glasses are recorded wearing glasses and without glasses two group data. When the recorded object is placed in different positions, it has different photosensitive effects, and the imaging is also different. This phenomenon is particularly pronounced for printed photos. Therefore, when we collect image pairs for print attack attempts, we move the recorded photo both vertically and horizontally besides moving back and forth to record images in different illumination condition. We record 500 frame pairs for every real access and attack attempt. The frames are saved as 600*480 resolutions in session 1 and session 2 but 1280*960 resolutions in session 3.

3.3. Data pre-processing of CIGIT-PPM

The background scene of every session is different. In order to exclude the possibility of using too much background information for face anti-spoofing and make the proposed dataset more challenging, we remove the background in the original frames and provide aligned face images. We use face detector introduced in [27] to detect face region from the collected VIS and NIR frames. All detected faces are normalized to 128*128. Sample aligned VIS and NIR face image pairs of CIGIT-PPM dataset are shown in Fig. 5. The aligned face image pairs are divided into training, development and testing set. There is no overlap for subjects in each set. The training, development and testing sets have 44, 14 and 14 subjects, respectively. In order to maintain the balance of positive and negative samples, we randomly remove some face image pairs of print attack. The detail data of face image pairs consisted in training, development and testing sets is shown in Table 2.

Table 2

The number of VIS and NIR face image pairs per type of sample.

	Train	Dev	Test	Total
Real access	28,350	9714	9715	47,779
Print attack	19,504	6495	6495	32,494
3D Mask attack	8834	2125	2126	13,085
Total	56,688	18,334	18,336	93,358

4. Proposed method

The main idea of the proposed method is to combine the advantages of both VIS image and NIR image for the goal of using a single model to distinguish print and 3D mask attacks from real accesses. As shown in Fig. 6, we employ convolution neural networks to learn fusing representation from paired VIS and NIR images at data level, convolution level and full-connected level, respectively. Then the multilevel classification scores are integrated as the final live and spoof face prediction result via weighted averages computation.

4.1. Paired VIS and NIR face images fusing

There are many motivations to fuse the features of paired VIS and NIR face images. VIS face images have been widely used in various previous face anti-spoofing methods. VIS face images contain rich cues for face anti-spoofing. For example, the secondary imaging of print attacks leads to the presence of image blur and color distortion in spoof face images. The color of the spoof face producers (e.g. printers, manufacturing plants of the masks) is limit compared to live faces, which also results in color distortion of spoof face images. The manufacturing processes of spoof faces cause that many spoof faces suffer from the problems like printing defects and lack of sufficient details. In addition, the paper surface leads to specular reflection in spoof face images. It is benefit and necessary to utilize the cues in VIS face images for face anti-spoofing. VIS face image are captured by visible light spectral camera. The spectral bands are just human eyes can apperceive. In other word, VIS face images are recorded as human eyes see. However, the final goal of making a successful spoof face is fooling human eyes and brain, so a high quality spoof face will try to evade the fake characteristics that can be found by human eyes under visible light. Hence only using VIS face images for anti-spoofing is not enough at all.

There are obvious reflection differences between common materials used to make spoof faces and the materials of live faces under near-infrared spectrum. These differences are all preserved in NIR images of live and spoof faces. As shown in Fig. 5, we can see that skin of live faces looks more smooth, delicate, elastic and shiny than spoof faces. The eyes of live faces look a little turbid, but the eyes of spoof faces look either clear or lack of details. The nose tip, eyeballs, teeth, glasses, and forehead of live and spoof faces have different specular reflection. These characteristics are all important cues we expect to utilize to improve the performance of face anti-spoofing. For theses cues, VIS face images are less obvious than NIR face image, but the NIR face images are black-and-white images, which do not preserve rich color and texture information like VIS face images. Following this perspective, VIS and NIR face images can complement each other for face anti-spoofing. Fusing the cues from these two types of images to improve performance of face anti-spoofing is worth expecting.

4.2. CNN-based multilevel fusing and classification

Considering the ability of CNN as well as the data volume, variations of our collected dataset, we utilize CNN to automatically

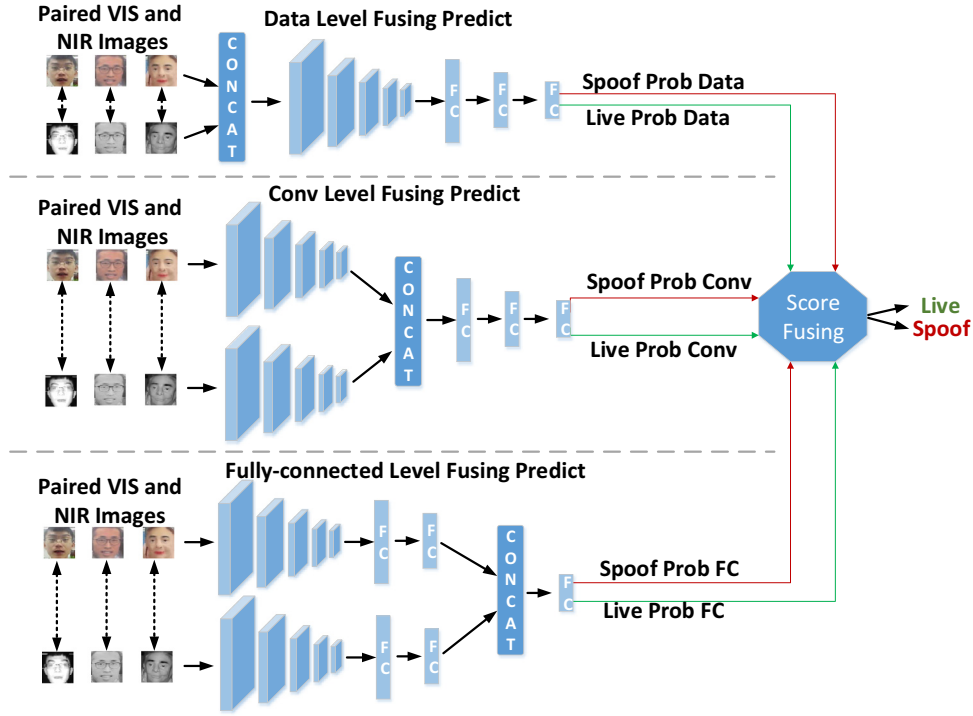


Fig. 6. Architecture of the proposed face anti-spoofing approach based on paired VIS and NIR face images fusing.

learn fusing features from paired VIS and NIR face images for face anti-spoofing. We regard the face anti-spoofing problem as a binary classification problem. Whether the presentation attack is a print attack or a 3D mask attack, we all classify it as spoof face. We utilize aligned VIS and NIR face image pair as the input of CNN model. The CNN model outputs the live and spoof probability for that paired face images. Different layers in convolution neural network learn different information from the input images. Lower layers capture the texture and structure information. Higher layers capture the semantic content information. For mining the cues in the paired VIS and NIR face images, we learn multilevel fusing representations of paired images at data level, convolution level and full-connected level.

An end-to-end convolution neural network is proposed to learn corresponding fusing representation and predict the live and spoof face classification probability at different network level. The network structure consists of two branches, one branch for learning features from VIS face image and the other for learning features from NIR face image. We adopt the network structure in [2] as the backbone of every branch, but it can be replaced by other network structure. As shown in Fig. 6, the network mainly consists of one data layer, five convolution layers and three fully-connected layers. To learn the fusing features of VIS and NIR face images, the two branches are concatenated at different layers. For data level fusing, we concatenate the paired VIS and NIR face images at data channel and feed it to the following network. For convolution level fusing, the VIS and NIR branches are concatenated after 5 convolution layers. For fully-connected level fusing, we concatenate the VIS and NIR branches at penultimate full connected layer and feed the fusing representation to the last full-connected layer and Soft-max layer.

In addition to the above trunk structure, we adopt ReLU and max pooling layer for each convolution layer. Batch normalization is used for every layer except the last fully-connect layer. Dropout is used after the first fully-connected layer to avoid over-fitting. The dropout value is 0.5.

Table 3

The protocol of msspoof [6].

	Spectrum	Train	Dev	Test	Total
Real access	VIS	315	210	210	735
Real access	NIR	315	210	210	735
Print attack	VIS	648	432	432	1512
Print attack	NIR	648	432	432	1512

Representations of different level have different priorities in their own classification process, so we utilize the score fusing strategy to integrate the results of different level. Classification probabilities of multiple level representations are weighted averaged to get the final prediction of the whole approach according to formula 1. p_{data} , p_{conv} , p_{ip} are the live and spoof prediction probabilities of data, convolution and fully-connected layers based branch concatenating networks, respectively. w_1 , w_2 , w_3 are corresponding weight gained from experiments. p is the final prediction, live and spoof probabilities of the paired VIS and NIR face images. Paired face images are classified as live faces if the final live probability is greater than threshold gained from experiments.

$$p = w_1 * p_{data} + w_2 * p_{conv} + w_3 * p_{ip} \quad (1)$$

5. Experiments

5.1. Experimental setup

Dataset We evaluate the effectiveness and generalization of our method via intra-dataset testing on CIGIT-PPM and msspoof datasets, as well as cross-dataset testing between them. Our experiments follow the protocol shown in the Tables 2 and 3 on CIGIT-PPM and msspoof, respectively. The attacks recorded in msspoof include VIS and NIR spoofing attacks. The images of print attack in Table 3 under VIS or NIR spectrum all consist of the images of VIS spoofing attacks and NIR spoofing attacks.

Table 4

Performance comparison of different level fusing for face anti-spoofing mixed with print and 3D mask attacks on CIGIT-PPM dataset.

Fusing level	EER(%)	HTER(%)
Data level	1.31	1.21
Convolution level	0.64	0.66
Fully-connected level	0.50	0.49
Score Fusing result	0.45	0.44

Experimental setting Firstly, we detect faces in the VIS and NIR frame pairs via face detector in [27]. Then, the paired face images are all resized to 96*96 as the input of the CNN model. All the input images are preprocessed by mean subtraction. We use Caffe toolbox [9] to train and test our model. The setting of training our model are listed as follow: The learning rate is 0.001, the momentum is 0.99, the weight decay is 0.0001 and the batch size is 100. We use the backbone network structure to train the classification model when the input image is single type image.

Evaluation metrics Because the cost of the mask is high, the subjects of mask attack is less than print attack and real access. A subject-exclusive five folds cross validation strategy is utilized to evaluate the performance for 3D mask attack. We use Equal Error Rate (EER) and Half Total Error Rate (HTER) [5] to evaluate the performance of different methods.

5.2. Multiple level fusing analysis

In this section, we evaluate the components of our proposed network, which fuses features at different feature levels. The EER and HTER results on the CIGIT-PPM dataset are shown in Table 4. We can find that the performance is improved step by step with the depth of the concatenation level increasing. Fusing the paired VIS and NIR face image features at fully-connected level achieves the best performance, which has an increase of 59.5% on HTER compared with data layer concatenation. Relative to concatenation at other levels, the interplay between VIS images and NIR images is less when we do concatenation at fully-connected level. It reflects that the cues in VIS and NIR face images can complement each other for face anti-spoofing. We utilize the weighted average of multilevel classification scores as the final score of our proposed method, where the weights are experimentally determined to be 0.3, 0.6 and 0.8 for data level, convolution level and full connected level, respectively. The data provided in Table 4 shows that integrating multilevel fusing scores can get an increase of 10% compared to fully-connected level fusing. This indicates that although the EER and HTER of the fully-connected level fusing are much smaller than the levels in front, every level has its own strength. Lower level still can call back some samples which are classified wrong by fully-connected level fusing. Integrating multilevel fusing results improves the performance of the whole method.

5.3. Paired images fusing analysis

The proposed approach is designed to fuse the cues in paired VIS and NIR face images, so we do experiments to explore the performance comparison of single type input images and the paired input images. We use VIS images, NIR images and paired VIS and NIR face images as input to train and test model, respectively. Table 5 shows the comparison results in terms of the EER(%) and HTER(%) for print attack, 3D mask attack, and presentation attacks mixed with print and 3D mask attack. From Table 5, we can see that the HTER of NIR image as input increases 68.5% compared to VIS images as input for print attack. Due to the materials and two-dimensional flat structure of printed photos, the NIR

Table 5

Performance comparison of single type images and the paired images as input for different presentation attacks on CIGIT-PPM dataset.

Presentation attack	Input image	EER(%)	HTER(%)
Print	VIS	5.09	4.90
Print	NIR	1.43	1.54
Print	VIS and NIR	0.12	0.23
3D Mask	VIS	5.51	5.56
3D Mask	NIR	11.2	11.11
3D Mask	VIS and NIR	0.95	0.87
Print and 3D mask	VIS	2.39	2.42
Print and 3D mask	NIR	1.68	1.57
Print and 3D mask	VIS and NIR	0.45	0.44

images of print attacks are significantly different from that of live faces. Though VIS images provide rich color and texture for the CNN model, NIR image is better at recognizing print attack. However, Mask has three-dimensional structure like live human beings. Compared to printed photos, the near-infrared characteristics of 3D masks and live faces are more similar. In this case, VIS images containing rich color and texture are more effective to recognize 3D mask attack than NIR images, achieving higher HTER of 49.9%. Generally, the VIS images and NIR images both contain important cues for face anti-spoofing. They have their own merits and uses.

From Table 5, we can also see that whether distinguishing one type presentation attack or two type mixed presentation attacks, paired VIS and NIR face images as the input can get significant performance improvement compared to single type images as input. Compared to the better one of the two single type images, HTER of paired images as input is 85%, 84.3%, 72.6% lower than the single type images for print attack, 3D mask attack and print-3D mask mixed attacks, respectively. The EER and HTER sharply go down when we use paired images as input. The figures in Table 5 lead us to the conclusion that the cues in VIS and NIR face images not only are differently qualified but also can complement each other. The paired images fusing helps the classification accuracy to be higher.

5.4. Mixed presentation attacks analysis

From another angle, we can also see the generalization of the proposed method from Table 5. Table 5 shows that the proposed VIS and NIR images fusing method can all get higher performance to detect print attack as well as 3D mask attack. Furthermore, it is effective not only for one type of presentation attack but also for two type mixed presentation attacks. The differences between print attack and 3D mask attack are larger than that between print attack and display attack. Many previous methods which utilize some particular cues such as random specified actions, 2D flat structure, edges of the printed photos or screens, motion information are effective to distinguish print and display attacks from real accesses. However, the performance of these methods will degrade because the exploited cues are not conducive to distinguish 3D mask attacks from real accesses. The experiment results indicate that the proposed model has the ability to learn generalized representation for the present common presentation attacks. For real world scenarios, print attack and 3D mask attack often occur mixed. A method which can handle mixed attacks is able to better meet the demand of face recognition systems.

5.5. Intra-dataset testing

We compare the performance of intra-dataset testing on CIGIT-PPM and msspoof datasets. We utilize all face image pairs of CIGIT-PPM for the experiments, which include both print and 3D mask

Table 6
Intra-dataset testing results on CIGIT-PPM and msspoof datasets.

Method	Input image	CIGIT-PPM		msspoof	
		EER(%)	HTER(%)	EER(%)	HTER(%)
LBP [11]	VIS	17.85	17.56	16.79	13.08
LBP [11]	NIR	8.59	8.64	10.59	16.73
Gabor [26]	VIS	4.40	4.32	13.03	22.67
Gabor [26]	NIR	7.10	7.25	18.99	29.19
CNN [24]	VIS	5.74	5.46	17.31	24.41
CNN [24]	NIR	1.51	1.69	12.39	15.75
PatchCNN [2]	VIS	1.70	1.57	10.95	13.81
PatchCNN [2]	NIR	1.61	1.73	15.59	9.48
Proposed method	VIS & NIR	0.45	0.44	9.37	3.43

Table 7

Cross-dataset testing results between CIGIT-PPM and msspoof. The evaluation metric is HTER(%). CM refers to using CIGIT-PPM for training and msspoof for testing. MC refers to using msspoof for training and CIGIT-PPM for testing.

Method	Input image	CM	MC
LBP [11]	VIS	48.83	49.45
LBP [11]	NIR	48.55	33.67
CNN [24]	VIS	49.79	40.31
CNN [24]	NIR	36.58	36.10
PatchCNN [2]	VIS	39.79	42.41
PatchCNN [2]	NIR	35.37	17.42
Proposed method	VIS and NIR	29.75	10.23

attacks. We evaluate the performance of existing methods using NIR image and VIS image as input, respectively. Table 6 shows the comparison results. From Table 6, we can see that our proposed method achieves a HTER of 73.96% and 63.81% improvement over the state-of-the-art method on CIGIT-PPM and msspoof datasets, respectively. Our network structure and training parameters of convolution neural network are the same as PatchCNN [2]. Therefore, the results indicate that extracting the fusing features from paired VIS and NIR images can get more discriminative features for face anti-spoofing. However, it is worth mentioning that deep learning based methods do not obtain remarkable performance improvements compared to LBP based method on msspoof. We think an important reason lies in that the data volume of msspoof is too small for deep learning based methods to learn discriminative features. Therefore, it is necessary to collect new datasets which provide more live and spoof face images for the development of face anti-spoofing.

5.6. Cross-dataset testing

We evaluate the generalization of proposed method via cross-dataset testing between CIGIT-PPM and msspoof. We utilize protocol CM to represent the cross-dataset testing which is training on CIGIT-PPM and testing on msspoof, and protocol MC to represent the cross-dataset testing which is training on msspoof and testing on CIGIT-PPM. Table 7 compares the cross-dataset testing results of different methods in terms of HTER. Our proposed method reduces a HTER of 15.88% and 41.27% compared to previous state-of-the-art method on protocol CM and MC, respectively. The improvement of performance on cross-dataset testing demonstrates the generalization and superiority of our proposed method. The VIS and NIR image fusing is effective to improve the generalization for face anti-spoofing.

However, it is worth noting that the HTER of our method on protocol CM is higher than the HTER on protocol MC in Table 7. Examples of failure cases of cross-dataset testing on protocol CM are shown in Fig. 7. Many failure images are too dark to clearly see



Fig. 7. Four failure examples of cross-dataset testing on protocol CM. The first two columns are live faces and the other two are spoof faces. The first row are VIS images and the second row are NIR images.

whole face. This is because the images in msspoof are recorded in several designed illumination conditions, like controlled ambient illumination with blinds down, no illumination. As a consequence, the illumination conditions of CIGIT-PPM are less complex than those of msspoof. Therefore, it is difficult for the model trained on CIGIT-PPM to correctly classify these images. This indicates the future research direction of enriching our CIGIT-PPM dataset.

6. Conclusion

In this paper, we collect a novel dataset named as CIGIT-PPM which contains paired visible light and near-infrared spectral images for both print and 3D mask attacks. Compared to prior similar public datasets, CIGIT-PPM has more data volume and diversity. Then, we introduce a novel face anti-spoofing method to utilize the cues in both visible light and near-infrared spectral images. First, we utilize an end-to-end CNN to learn fusing representation from paired VIS and NIR images and do classification at different feature levels. Then multilevel classification results are integrated to distinguish the live faces from spoof faces. The experiment results demonstrate the superiority of proposed method.

Declaration of competing interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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